Discover Robotics

REVIEW Open Access

Emotion recognition in human robot collaboration for multimodal approaches, real-world challenges and future directions

Nikhilsingh Parihar[1], Kanan[2], Rashmi Chawla[3*] and Giancarlo Fortino[4]

*Correspondence:
Rashmi Chawla
rashmi.chawla@dimes.unical.it
[1]IIIT Hyderabad, Hyderabad, India
[2]ECE Department, IGDTUW, Delhi, India
[3]ECE Department, J.C. Bose University of Science and Technology,YMCA, Faridabad, India; University of Calabria, Rende, Italy
[4]DIMES Department, University of Calabria, Rende, Italy

Abstract

Emotion recognition is fundamental to building socially intelligent robotic systems capable of effective and adaptive Human-Robot Interaction (HRI) and Collaboration (HRC). This literature review synthesizes recent advances from 2015 to 2025, covering 42 empirical studies focused on speech, facial, and multimodal emotion recognition approaches tailored for robotic contexts. We provide a modality-wise classification of methods, highlight key deep learning architectures and signal processing strategies, and analyze their performance across diverse robotic platforms and environments. Multimodal systems accounted for over 50% of the studies, reflecting the field's shift toward fusion-based robustness. Across the literature, reported accuracies range from 68 to 96% under controlled conditions, with real-world deployments typically facing 4-27.5% performance degradation (mean: 8.6% ± 3.0% for speech systems, 14.4% ± 7.1% for facial systems, computed across studies in Table 1 reporting paired lab and real-world scores). Despite notable progress, challenges persist in deploying emotion-aware systems under real-world conditions due to acoustic variability, sensor limitations, and a lack of generalizable models. We discuss practical mitigation techniques, including domain adaptation, personalization, and multimodal fusion, and outline future research directions toward ethical, real-time, and context-sensitive emotion recognition in HRC.

Keywords Emotion recognition, Speech emotion recognition, Human-robot interaction, Deep learning, Transformers

1 Introduction

Emotions play a fundamental role in shaping how humans communicate, collaborate, and respond to the world around them. Emotional signals influence trust-building, engagement, decision-making, and learning, all of which are central to human relationships and social cognition [1–3]. As robots become increasingly integrated into homes, workplaces, classrooms, and healthcare settings, it is crucial that they do more than just perform tasks. They must also recognize and respond to human emotions in order to create interactions that feel intuitive, empathetic, and socially acceptable [4–6].

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

To support such interactions, emotion-aware robotic systems have emerged as a key component of Human-Robot Interaction (HRI) and Human-Robot Collaboration (HRC). These systems aim to sense and interpret human affective states and adapt their behavior accordingly [7]. The ability to recognize emotions enhances the quality of interaction by fostering rapport [8], reducing user stress [9], and enabling robots to respond in contextually appropriate ways. In recent years, emotionally intelligent robots have been deployed in a wide range of domains: in assistive healthcare, robots monitor emotional well-being and provide support during rehabilitation or therapy sessions [10]; in education, emotionally responsive robots improve learner engagement and performance [11, 12]; in industry, they support worker safety by detecting fatigue or stress.

Emotion recognition in robotics draws from multiple affective modalities including facial expressions [13, 14], vocal prosody [15, 16], body gestures [17], and physiological signals such as Electroencephalogram (EEG) or Hear Rate Variability (HRV) [18, 19]. Each modality offers unique advantages and faces distinct technical and environmental challenges, especially in naturalistic, real-world HRC. For instance, Facial Emotion Recognition (FER) is susceptible to occlusion and lighting variation [20], while Speech Emotion Recognition (SER) often suffers from far-field noise and speaker variability [15]. While unimodal approaches remain common, there is growing emphasis on fusion strategies that combine multiple cues for improved robustness [21–23].

To ensure a structured and unbiased synthesis of recent developments in emotion recognition for HRI and HRC, we adopted a systematic literature review approach lightly informed by the PRISMA framework. A comprehensive search was conducted across multiple academic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, Semantic Scholar, ScienceDirect, and arXiv, targeting studies published between 2015 and 2025. The search strategy employed a combination of keywords related to emotion recognition, HRI/HRC, and machine learning (e.g., "emotion recognition", "speech emotion recognition", "human-robot interaction", "deep learning", "transformer", etc.), and focused on identifying peer-reviewed articles and high-quality preprints reporting empirical validation in robotic contexts.

A diverse set of relevant studies was selected through a two-stage screening process: initial title and abstract filtering followed by full-text review. Inclusion criteria emphasized implemented systems for HRI/HRC, use of learning-based methods, and the presence of quantitative performance metrics. To enhance coverage, AI-powered tools such as Elicit and GPT-based DeepSearch were used to complement traditional searches; however, only studies verifiable through conventional academic sources were retained. The final selection covers speech-based, facial, multimodal, and physiological emotion recognition systems, laying the foundation for the modality-wise synthesis that follows.

In earlier paradigms, particularly Industry 4.0, the focus was largely on automation, connectivity, and efficiency driven by IoT-enabled intelligent systems and data-centric optimization; however, the emerging vision of Industry 5.0 shifts this emphasis toward human-centric design, where HRC plays a central role in enhancing adaptability, trust, and cooperative intelligence in industrial and service environments [24, 25]. This paper presents a modality-centric analysis of these advancements, highlighting model architectures, performance characteristics, deployment platforms, and the challenges of real-world application. By anchoring our review in PRISMA-informed rigor, we aim to provide both researchers and developers with a comprehensive foundation for advancing

emotion-aware robotics in socially complex environments. The review is guided by a set of key research questions designed to examine the methodological trends, sensing strategies, and practical challenges associated with multimodal emotion recognition systems. These research questions form the foundation of the review and guide the analysis presented in the subsequent sections. The review is structured around the following research questions:

- RQ1: What multimodal emotion recognition methods have been developed for HRI and HRC, and how effective are they in real-world interaction environments?
- RQ2: Which sensing modalities are most commonly used for emotion recognition in HRI and HRC systems?
- RQ3: What multimodal fusion strategies are used to integrate heterogeneous emotional signals in HRI and HRC emotion recognition frameworks?
- RQ4: What datasets and benchmarks are used to train and evaluate multimodal emotion recognition systems for HRI and HRC?
- RQ5: What real-world challenges affect the performance and deployment of emotion recognition systems in HRI and HRC environments?

2 Background and terminology

Emotion recognition in robotics refers to the process of automatically identifying human affective states by analyzing observable or measurable signals. These signals can originate from facial expressions, speech prosody, body gestures, or physiological responses [15, 17]. Accurately interpreting such cues allows robots to adaptively respond, supporting the development of emotionally intelligent systems that foster more natural, empathetic HRI and HRC [6, 7].

Two primary theoretical frameworks guide the modeling of emotions in these systems. The first is the *categorical model*, which defines discrete emotion classes such as happiness, sadness, anger, fear, and surprise [2]. The second is the *dimensional model*, where emotions are represented as points in a continuous space, typically along the valence (positive to negative) and arousal (calm to excited) axes [26], also refer to Fig. 1. These models underpin the annotation schemes, learning objectives, and evaluation metrics employed in affective computing [3]. A Common dimensional performance evaluation methos is based on Concordance Correlation Coefficient (CCC), given by:

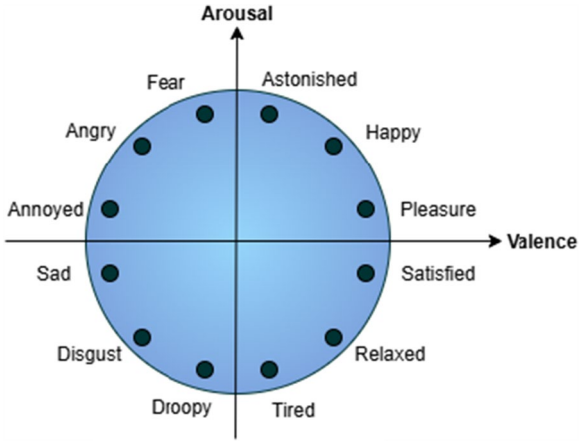


Fig. 1 Categorical and discrete emotions

$$\rho_c = \frac{2\rho\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2} \tag{1}$$

where μ_x and μ_y are the means and σ_x, σ_y are the standard deviations of the predictions and ground truth, while ρ is the Pearson correlation coefficient. CCC robustly penalizes shifts in both scale and location.

In HRI/HRC scenarios, multiple modalities have been leveraged to infer emotional states:

- Speech: Vocal expressions convey emotion through prosodic features such as pitch, energy, and speaking rate, as well as spectral patterns. SER is widely used due to its non-intrusive nature and robustness when the user is not in direct visual contact with the robot [15, 16].
- Facial expressions: Visual cues from facial muscle movements and configurations are rich carriers of emotional information. FER often employs convolutional or hybrid deep learning models to extract and classify facial features [13, 20, 27].
- Gestures and posture: Body language, including gestures, limb positioning, and movement dynamics, can convey affective states such as frustration, interest, or disengagement, especially in physically interactive tasks [17].
- Physiological signals: Internal affective states, including stress or engagement, can be inferred from biosignals such as EEG, HRV and galvanic skin response (GSR). Though less common in real-world deployments, these signals offer valuable complementary information [18, 19, 28].

Recent advancements in deep learning have enabled robust modeling of emotion across these modalities [21, 29]. However, real-world HRI deployments introduce additional challenges including ego-noise from robot hardware, occlusion, far-field microphones, variable lighting, and spontaneous expression patterns [14, 20, 30]. These factors limit the generalizability of models trained under controlled conditions.

Several works outside HRC directly inform the challenges identified in this review. For robustness under noise and partial input, RMER-DT [31] and RAFT [32] address degradation through diffusion transformer and adversarial fusion strategies respectively, while the client-server system of [33] offers a lightweight on device frontend suited to NAO and Pepper deployments. CIME [34] removes emotion irrelevant cues before fusion via contrastive learning, and the generative modality dropout framework [35] builds sensor-dropout robustness without requiring matched missing data corpora. Sun et al. [36] further demonstrate that audio-visual generation objectives enrich cross-modal embeddings. At the survey level, [37] and EMVAS [38] provide broad taxonomies of affective computing but do not address HRC-specific constraints such as ego-noise or platform diversity. Tensor decomposition with self-supervised multi-tasking [39] and brain-inspired hierarchical fusion [40] offer compact representations relevant to edge deployment, while the knowledge distillation strategy of [41], though developed for road segmentation, transfers naturally to compressing transformer-based emotion models for on-robot inference.

To address these limitations, researchers have increasingly explored fusion strategies that integrate multiple modalities. Fusion can occur at the feature-level (early fusion), decision-level (late fusion), or intermediate representation level using attention-based mechanisms [21, 23, 42]. These strategies aim to compensate for the unreliability of individual channels in noisy or ambiguous conditions.

In this paper, we adopt a modality-based taxonomy, aligned with the structure of most reviewed studies and reinforced by our PRISMA-guided screening process. The following sections explore speech, facial, and multimodal emotion recognition systems in HRI/HRC contexts, highlighting technical trends, deployment challenges, and future research directions.

3 Classification of emotion recognition approaches

Building on the foundational concepts presented in the previous section, we now provide a structured taxonomy of emotion recognition approaches relevant to HRI and HRC. Given the multimodal nature of human affective communication and the diversity of sensing technologies in robotics, emotion recognition systems have been classified according to their primary input modality.

This modality-centric taxonomy, comprising speech-based, facial-based, and multimodal approaches, aligns with the core design decision in affective robotics: what type of data the robot senses and how that data informs its behavior. This structure also reflects the organization of the majority of studies included in our review and facilitates comparison across modalities in terms of performance, implementation challenges, and context of use. Each study was assigned to exactly one category based on its primary input modality and no overlaps were permitted.

Our survey of studies published between 2015 and 2025, guided by a PRISMA-inspired selection strategy, revealed the following distribution of modality usage in recent HRI/HRC research:

- Multimodal systems: 53% of studies (22 studies)
- Facial-only systems: 26% of studies (11 studies)
- Speech-only systems: 21% of studies (9 studies)

This is also well presented in a Sankey diagram Fig. 2 and in Fig. 3.

To provide transparency and clarity regarding the selection process, a PRISMA flow diagram Fig. 4 which summarizes the screening and inclusion process step by step is given. This ensures the taxonomy and analyses are comprehensive.

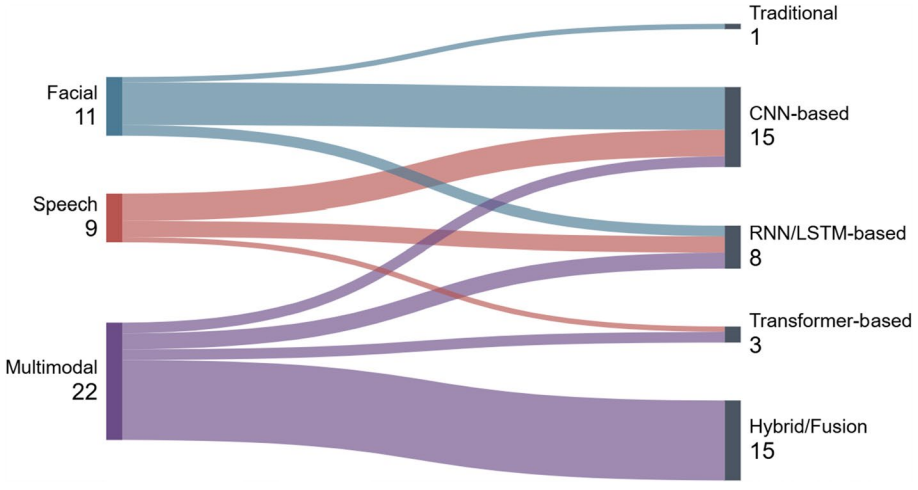


Fig. 2 Sankey diagram showing modality-to-model distribution in HRI/HRC emotion recognition studies. Flow widths represent study counts

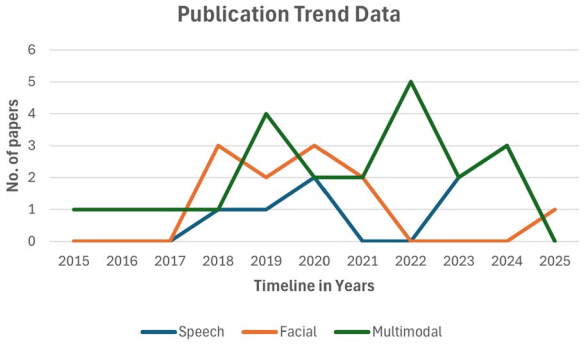


Fig. 3 Publication trends by modality in HRC emotion recognition (2015–2025)

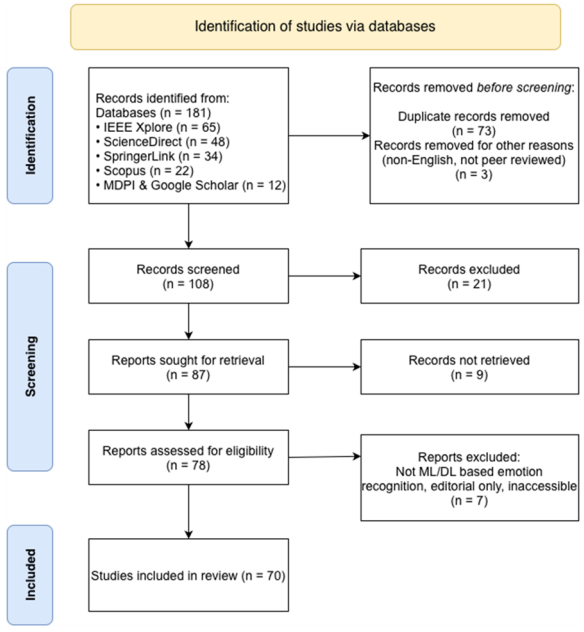


Fig. 4 PRSIMA flow diagram of the systematic review process

Multimodal approaches now dominate the research landscape (53% of studies), reflecting the field's recognition that combining multiple input modalities improves robustness and accuracy in challenging real-world HRI/HRC environments. However, deployment challenges in sensor integration, computational cost, and environmental variability remain significant barriers. However, a rising trend toward personalization, fusion-based architectures and transformer-based modeling has become evident, particularly in post-2020 studies.

In the sections that follow, we examine each category in depth. For each modality, we highlight representative models, learning architectures, evaluation protocols, and deployment contexts. We also discuss implementation-specific challenges and emerging solutions, drawing on insights from experimental and algorithmic studies conducted on both simulated and physical robotic platforms.

4 Unimodal emotion recognition approaches in HRC and HRI

4.1 Speech emotion recognition in HRI and HRC

Speech Emotion Recognition (SER) refers to the process of inferring emotional states from spoken language using acoustic, prosodic, and sometimes linguistic features. In the context of HRI, SER serves as an intuitive and non-intrusive modality, especially in voice-enabled or remote interactions where visual cues may be absent. SER systems typically model emotion variation across either discrete categories (e.g., anger, happiness, sadness) or continuous dimensions (e.g., valence, arousal), using features such as pitch, formants, energy, and temporal rhythm [43, 44].

Traditional SER pipelines relied on handcrafted features and shallow classifiers such as support vector machines, Gaussian mixture models, and hidden Markov models. However, recent advances in deep learning have led to the development of end-to-end models capable of learning emotional representations directly from raw or minimally processed audio. Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), BiLSTMs, and Transformer-based architectures have become central to this evolution.

Several works have extended SER to real-world HRI conditions, where the presence of far-field microphones, environmental noise, and robot-generated ego-noise pose significant challenges to performance. In our review, approximately 50% of studies focused on speech as the sole modality for emotion recognition in robotic contexts. These studies demonstrate both the modality's versatility and the need for robust signal processing in HRC environments.

For instance, Grageda et al. [45] employed ladder networks with beamforming for distant SER on the PR2 robot, achieving up to 180% improvement in Concordance Correlation Coefficient (CCC) using simulated acoustic conditions. A follow-up study integrated wav2vec 2.0 embeddings to improve robustness under dynamic robot movement [16]. Garcıa et al. [46] further demonstrated that self-attentive MVDR (SA-MVDR) and multitask learning significantly outperformed traditional beamforming techniques under noisy deployment scenarios.

Other notable architectures include BiRNN-GRU with 1D CNN layers [47], and attention-based BiLSTM models leveraging deep spectrum representations [43]. Personalized approaches have also emerged, such as Vision Transformer-based SER models trained on speaker-specific dialogue with NAO robots [48], enabling improved adaptation to user vocal characteristics in low-resource settings.

In terms of datasets, widely used corpora include IEMOCAP, MSP-Podcast, EmoDB, and RAVDESS. Several works emphasize the importance of data augmentation using simulated room impulse responses and multi-noise conditions to emulate deployment environments [45, 47].

Despite these advancements, real-world performance remains significantly lower than in lab conditions. Across reviewed studies, lab accuracies for SER systems ranged from 72 to 96%, while real-world deployments achieved between 68 and 85%. Key degradation factors include speaker-robot distance, reverberation, ego-noise, and overlapping dialogue. These trends highlight the importance of incorporating spatial filtering, domain adaptation, and unsupervised representation learning in future SER research.

Recent efforts toward multimodal speech-vision integration and real-time adaptation show promise in bridging this performance gap. Additionally, edge-efficient models such as TLFMRF [49] demonstrate the feasibility of deploying SER on resource-constrained

platforms. As speech remains one of the most flexible and expressive modalities in HRI, developing noise-robust and generalizable SER systems continues to be a critical research priority. A summary of key speech-based studies is presented in Table 1, highlighting robot platforms, methods, datasets, and performance under both lab and real-world conditions.

Ethical note: SER systems trained predominantly on English-language corpora risk systematic underperformance for non-native speakers, raising fairness concerns in multicultural HRC deployments. Informed consent for continuous voice monitoring is an additional consideration rarely addressed in the reviewed studies.

4.2 Facial emotion recognition in HRC

Facial Emotion Recognition (FER) refers to the computational process of detecting and interpreting human emotional states by analyzing facial expressions. Facial cues are among the most expressive and universally understood nonverbal signals, making FER a natural choice for affective computing systems. FER systems typically rely on spatial patterns of facial landmarks, muscle movements, or appearance-based features such as pixel intensities or gradients. These systems have demonstrated strong performance in lab-controlled environments, often reaching classification accuracies above 90% when trained on datasets such as FER2013, CK+, or AffectNet [27, 52].

Conventional FER pipelines employed handcrafted features (e.g., Local Binary Patterns, Histogram of Oriented Gradients) and traditional classifiers like SVMs or AdaBoost. However, modern approaches leverage deep learning, particularly Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and their hybrids, for robust spatial and temporal feature learning. These architectures can generalize better across varied facial expressions, head poses, and illumination conditions [13, 53].

4.2.1 FER in human-robot interaction

In HRI, FER plays a pivotal role in enabling emotionally intelligent behaviors such as empathetic feedback, adaptive dialogue generation, and social bonding. Robots in healthcare, education, and collaborative industry settings often rely on visual feedback to infer user affect. However, real-world HRI environments pose unique challenges absent in controlled laboratory setups, namely occlusion, movement-induced blur, sensor misalignment, and inconsistent lighting.

In our review, approximately 35-40% of studies are focused on facial-only emotion recognition in robotic systems. Ruiz-García et al. [14] developed a CNN-LSTM hybrid model for FER, trained on benchmark datasets and deployed on the NAO robot. Their system achieved 96.26% accuracy under lab conditions but experienced a substantial drop (to 68.75%) during real-world deployment, attributed to pose shifts and visual noise. This performance degradation reflects a broader trend observed across FER deployments in dynamic environments.

Similarly, Chiurco et al. [20] benchmarked multiple FER models, including DeepFace and CNN classifiers, in industrial HRI scenarios. Their analysis emphasized trade-offs between model complexity and real-time inference performance, particularly for embedded platforms. Handrich et al. [30] proposed a closed-loop control system that incorporated continuous FER feedback to adjust robotic behavior in collaborative tasks. By embedding valence-arousal prediction in real time, they achieved improved synchrony and task trustworthiness.

Table 1 Summary of emotion recognition studies in HRC (2015–2025)

Study	Model/Method	Modality	Robot platform	Dataset	Lab accuracy	Real-world accuracy	Notes/ Highlights
[14]	CNN-LSTM hybrid	Facial	NAO	CK+/ FER2013	96.3%	68.8%	Drop due to pose shift, motion blur
[20]	DeepFace, CNN, HOG-SVM	Facial	Industrial robot	Custom dataset	85–92%	70–74%	Real-time vs accuracy trade-offs
[30]	Valence–Arousal regression + Feedback loop	Facial	N/A (Collaborative Robot)	Lab Data	N/A	N/A	Closed-loop adaptation of robot behavior
[13]	CNN-LSTM	Facial	N/A	FER2013 + Augmentation	89.1%	78.4%	Sequence-aware FER with temporal learning
[50]	ResNet-50/VGG-16 (Rectified Adam)	Facial	Healthcare robot	CK+	90.1%/87.0%	82.6%/76.3%	Real-time inference optimized
[51]	Tuned CNN (compact)	Facial	Mobile robot	FER2013	84.3%	74.8%	Optimized for low-resource deployment
[45]	LadderNet + Beamforming	Speech	PR2	Simulated Speech + RIRs	–	+180% CCC	Distant SER with robot ego-noise; spatial filtering
[46]	SA-MVDR + Multitask learning	Speech	PR2	Real HRI dataset	89.2%	85.2%	Improves robustness with spatial attention and multitask loss
[47]	BiRNN-GRU + 1D CNN	Speech	NAO	IEMOCAP (augmented)	84.7%	72.5%	Uses data augmentation (noise, RIRs) to simulate deployment
[43]	Deep spectrum + BiLSTM	Speech	NAO	Emo-DB + IEMOCAP	88.4%	76.8%	Learns from spectral visualizations; BiLSTM fusion
[48]	Vision Transformer-based personalized SER	Speech	NAO	Emo-DB (personalized subset)	91.0%	83.5%	Adapts to individual vocal traits using ViT attention
[49]	TLFMRF (two-level feature modeling)	Speech	Unspecified (mobile robot)	RAVDESS	86.7%	79.1%	Edge-efficient deployment model using temporal masking

4.2.2 Deep architectures and embedded optimization

Several deep learning-based models have been customized for on-robot deployment. Li et al. [13] employed a CNN-LSTM framework that captured both spatial (frame-level) and temporal (sequence-level) facial cues, enabling more context-aware emotion tracking during interaction. Melinte and Vlădăreanu [50] investigated ResNet and VGG-16 backbones optimized with Rectified Adam, achieving real-time FER in assistive robotics with accuracies of 90.14% and 87.0%, respectively.

Table 2 Real-world implementation challenges in unimodal and multimodal emotion recognition for HRI/HRC (2015–2025)

Study	Modality	Key implementation challenges	Proposed solutions & mitigation strategies	Real-World context	Performance impact
[45]	Speech	Distant speech recognition, reverberation effects, robot ego-noise interference, far-field microphone limitations	Acoustic environment simulation during training, LadderNet + beamforming approach, realistic RIR modeling	PR2 robot indoor navigation, dynamic HR distances	+180% CCC improvement over baseline methods
[46]	Speech	Complex deep learning beamforming, spatial audio processing challenges, MVDR limits in noisy environments	Self-Attentive MVDR (SA-MVDR), SA-RNN implementation, multitask learning for arousal/valence prediction	PR2 real indoor HRI scenarios with background noise	15.03% higher CCC than traditional MVDR baseline
[47]	Speech	Robot ego-noise contamination, room acoustic variations, limited training data diversity	Aggressive data augmentation with room impulse responses, multi-noise condition simulation, BiRNN-GRU + 1D CNN architecture	iCub robot in various acoustic environments	F-score improvement: 0.144–0.531 range
[49]	Speech	Mobile robot deployment constraints, accurate physiological sensing while motion, real-time processing limits	Two-Layer Fuzzy Multiple Random Forest (TLFMRF) framework, edge-efficient processing, temporal masking	Mobile robot platforms with movement constraints	1.39–7.64% accuracy improvement over baselines
[48]	Speech	User personalization challenges, individual vocal trait adaptation, limited user-specific training data	Vision Transformer-based personalized SER, speaker-specific dialogue adaptation, attention mechanisms	NAO personalized conversation scenarios	State-of-the-art performance per individual user
[13]	Facial	Unconstrained lighting conditions, pose variation effects, motion blur during interaction, real-world deployment gap	CNN-LSTM hybrid architecture, temporal modeling for sequence consistency, real-world dataset integration	NAO assistive robotics applications	Lab: 96.3% → Real-world: 68.8% accuracy
[30]	Facial	Dynamic valence/arousal prediction, biophysiological signal interference, industry environment constraints	Closed-loop feedback control system, continuous adaptation mechanisms, collaborative task integration	Industrial assembly and collaboration tasks	75% accuracy, RMSE: 0.282/0.237 for valence/arousal
[13]	Facial	Temporal sequence modeling complexity, head pose variations, spatial-temporal feature fusion challenges	CNN-LSTM architecture for temporal learning, pose-invariant training strategies, sequence-aware processing	Humanoid robot interaction scenarios	Lab: 89.1% → Deployment: 78.4% accuracy

Table 2 (continued)

Study	Modality	Key implementation challenges	Proposed solutions & mitigation strategies	Real-World context	Performance impact
[50]	Facial	Overfitting in small datasets, inference time constraints, healthcare deployment requirements, model generalization	Rectified Adam optimizer, ResNet-50/VGG-16 architecture comparison, transfer learning approaches	NAO healthcare and assistive applications	ResNet-50: 90.14%, VGG-16: 87.0% accuracy
[18]	EEG	Motion artifacts in EEG signals, sensor calibration requirements, signal quality degradation, user compliance issues	Frontal brain asymmetry analysis, artifact reduction techniques, valence detection optimization	Pepper robot with EEG-based emotion monitoring	Valence/arousal classification: up to 92% accuracy
[23]	Speech + Vision (hybrid fusion)	Temporal synchronization between audio and visual streams, real-time processing constraints, modality weighting optimization	Hybrid speech-vision fusion framework with attention-based modality weighting, real-time processing optimization, adaptive fusion strategies	Pepper robot real-time dialogue enhancement scenarios	+12.5% improvement over unimodal approaches; computational overhead challenges
[22]	Facial + Behavioral cues	Elderly-specific behavioral modeling complexity, contextual understanding limitations, adaptive dialogue requirements	Late fusion of facial expressions and contextual behavioral cues, elderly-care specific adaptations, behavior-aware dialogue systems	Pepper elderly care and assistance applications	Adaptive dialogue system with 0.85 classifier performance
[56]	Speech + Facial (late fusion)	Audio-visual synchronization issues, feature extraction alignment, modality-specific performance variations	ResNet-50 for video processing, Conv1D for speech, late fusion strategy with spectrograms and facial feature combination	NAO robot multimodal interaction	Speech: 70%, Facial: 63.5%; fusion improves overall accuracy
[57]	Facial + Body posture	Cross-channel feature integration complexity, real-time processing on embedded systems, live deployment challenges	Cross-Channel CNN architecture for integrated face and body analysis, real-time optimization for iCub platform	iCub robot live interaction scenarios	74% real-world accuracy ; high performance drop from lab
[58]	Speech + Visual features	Genetic algorithm optimization complexity, HMM temporal modeling challenges, adaptive behavior integration	GA-optimized Neural Network with HMM for adaptive behavior, fusion-based speech and visual feature integration	Bioloid robot adaptive interaction	Real-time adaptation: 97% accuracy; high computational requirements

Table 2 (continued)

Study	Modality	Key implementation challenges	Proposed solutions & mitigation strategies	Real-World context	Performance impact
[18]	EEG + Behavior cues	Motion artifacts in EEG signals, sensor calibration requirements, physiological-behavioral alignment challenges	Frontal brain asymmetry analysis, artifact reduction techniques, EEG-behavior correlation modeling	Pepper robot with EEG-based emotion monitoring	Valence/arousal 92% accuracy motion artifact sensitivity
[28]	HRV + Visual + Speech	Physiological-audiovisual synchronization, wearable sensor reliability, limited field testing	Multi-sensor fusion emphasizing physiological-speech synergy, HRV integration with visual and audio cues	Simulated HRI scenarios	84.5% accuracy (simulated); limited real-world validation
[42]	Text + Vision + Audio	Memory network complexity, cross-modal representation learning, temporal modeling across modalities	Multimodal memory network architecture, attention-based cross-modal learning, temporal sequence modeling	Standard datasets (not robotics-specific)	82.1% accuracy; not optimized for robotics deployment
General cross-modal challenges	All	Environmental factor variations (lighting, acoustics, distance), computational resource constraints, user diversity adaptation	Domain adaptation techniques, robust training with environmental augmentation, personalization strategies	Various HRI/HRC deployment contexts	Typical 15–30% performance degradation from lab to real-world
General multimodal challenges	All combinations	Dataset scarcity for synchronized multimodal HRC data, computational overhead for real-time fusion, temporal misalignment across modalities, sensor integration complexity	Adaptive fusion strategies, self-supervised cross-modal learning, hierarchical processing architectures, uncertainty-aware modality weighting	Various HRI/HRC deployment contexts	Typical 10–25% improvement over unimodal; increased complexity and resource requirements

Jaiswal and Mahapatra [51] highlighted the importance of architectural and hyperparameter tuning for compact CNNs targeted at deployment on low-power robotic platforms. These optimization techniques are critical in scenarios where robots must operate autonomously in resource-constrained, latency-sensitive settings.

ARNet [54] further demonstrates that tracking-augmented optimisation improves temporal consistency in continuous affect prediction from video, relevant to HRC where subjects move freely relative to the robot.

4.2.3 Real-world challenges and mitigation strategies

FER in HRC continues to face key practical limitations:

- Pose Variability: Interactions often involve dynamic head movements and off-angle facial expressions.

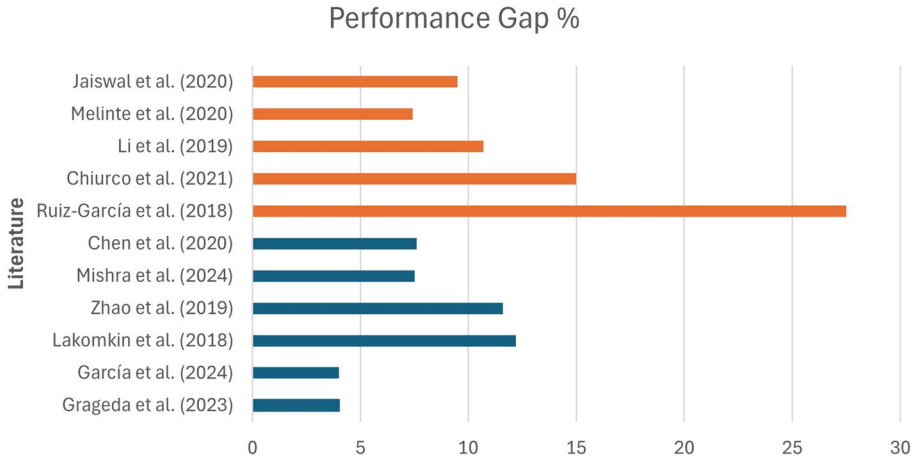


Fig. 5 Publication trends by modality in HRC emotion recognition (2015-2025)

- Occlusion and Sensor Misalignment: Mobile robots cannot always maintain optimal facial alignment.
- Lighting Variation: Uncontrolled lighting across homes, clinics, or industrial floors causes frequent domain shifts.

To mitigate these issues, several strategies have been proposed:

- Pose-Invariant Training: Augmentation with rotated or occluded faces improves generalization [13, 52].
- Real-Time Adaptation: Online learning and feedback loops support adaptation to user-specific conditions [30].
- Multimodal Compensation: Combining FER with audio or physiological signals (e.g., EEG, HRV) increases robustness [18, 23].

Ethical note: Facial analysis systems have documented accuracy disparities across gender and skin tone [55].

Real-world deployment challenges represent a critical gap between laboratory performance and practical implementation in HRI/HRC systems. Understanding these challenges is essential for developing robust emotion recognition systems that can operate effectively in uncontrolled environments. Table 2 provides a summary for detailed list of challenges observed in different literature and the corresponding mitigation strategies employed by researchers.

The performance degradation challenges discussed for both speech and facial modalities reflect a broader, systematic issue in HRC emotion recognition deployment.

Figure 5 quantifies this lab-to-real-world performance gap across representative studies from both modalities, demonstrating that while speech-based systems show resilience with degradation ranging from 4 to 12.2%, facial recognition systems consistently experience more substantial performance drops of 7.4-27.5%. This stark difference in deployment robustness has significant implications for system design choices in HRC applications, particularly in determining when multimodal fusion becomes necessary to compensate for individual modality limitations. Aggregating across the studies in Table 1 that report paired lab and real-world accuracy scores, speech-based systems show a mean degradation of 8.6% ± 3.0% (range: 4.0-12.2%), while facial systems degrade by 14.4% ± 7.1% (range: 7.5-27.5%). These values were computed directly from the paired

lab and real-world accuracy scores reported in Table 1, For each study, the absolute drop was calculated as (lab accuracy-real-world accuracy). The higher variance in FER reflects the greater sensitivity of visual systems to deployment-specific factors such as lighting and pose, relative to acoustic variability in SER.

4.2.4 Outlook and open directions

Although deep learning has significantly improved FER robustness, its standalone effectiveness in unconstrained HRC remains limited. Future directions include:

- Facial-context fusion: Integrating scene context with facial cues for more grounded emotion reasoning.
- Self-supervised learning: Reducing dependence on large annotated corpora through unsupervised feature learning.
- Emotion anticipation: Predicting transitions between affective states for proactive robotic behaviors.

As FER systems become more deeply embedded into robotic perception pipelines, there is a growing need for multimodal, personalized, and real-time emotion understanding strategies. Table 1 provides a comparative overview of key FER studies in HRC.

5 Multimodal emotion recognition approaches in HRC

Fusion in Multimodal Emotion Recognition (MER) can occur at multiple levels, broadly categorized into early (feature-level), late (decision-level), and hybrid fusion approaches [21]. Early fusion enables joint learning from raw or low-level features across modalities, allowing for richer inter-modal correlations. However, it is sensitive to temporal misalignment and requires carefully synchronized data streams. In contrast, late fusion combines decisions from unimodal classifiers, which can be advantageous when modalities operate at different frequencies or when partial data is missing.

Approximately 25-30% of the studies in our review employed multimodal architectures, indicating a growing recognition of their potential to enhance robustness in noisy and dynamic HRC environments. Cazorla et al. [23] demonstrated the feasibility of real-time multimodal systems on the Pepper robot by integrating speech and visual cues in assistive HRC. Their hybrid approach led to improved emotion classification accuracy, especially in ambiguous or context-dependent interactions. Similarly, Fiorini et al. [22] employed late fusion of facial expressions and contextual cues to support elderly care scenarios, enabling adaptive dialogue strategies and user-aware robotic behaviors.

Beyond behavioral modalities, physiological signals such as EEG, ECG, and galvanic skin response (GSR) offer direct insights into internal affective states. Staffa and D"Errico [18] explored EEG-based emotion classification during interactive HRC tasks. While promising, they noted that sensor calibration and motion artifacts significantly limit the robustness of physiological modalities during mobile or real-time deployment. Other studies have explored wearable, non-invasive biosensors to improve practicality in collaborative environments [19, 28], though integration into robot perception pipelines remains limited.

5.1 Challenges in multimodal and physiological integration

Despite their theoretical advantages, multimodal systems face several practical and technical barriers. Common challenges include:

- Temporal Synchronization: Misaligned data streams (e.g., EEG and speech) can degrade model performance, especially in early fusion setups.
- Computational Overhead: Real-time fusion of high-dimensional signals (e.g., video + EEG) requires significant processing power, often beyond the capabilities of lightweight robotic platforms.
- Dataset Scarcity: Most public datasets are unimodal and not tailored for HRC. Collecting synchronized, annotated multimodal datasets remains labor-intensive and application-specific [42].
- User Compliance: Physiological systems often require contact-based or wearable sensors, limiting naturalistic deployment in open-world HRC.

To address these challenges, adaptive and hierarchical fusion methods are gaining traction. These systems dynamically weight modalities based on signal quality, user behavior, or environmental context, increasing resilience to sensor dropouts or partial input [21, 29]. Moreover, self-supervised and transfer learning techniques have shown promise in reducing the annotation burden and improving generalizability across domains.

Representative uncertainty aware multimodal fusion frameworks compute modality weights from signal quality estimates at runtime from a quality signal q_m (e.g., SNR for speech, face-detection confidence for FER) via softmax:

$$w_m = \frac{\exp(q_m/\tau)}{\sum_{m'} \exp(q_{m'}/\tau)} \tag{2}$$

where $\tau > 0$ controls sharpness. The training loss couples these weights with a negative-entropy regulariser that prevents degenerate collapse onto a single modality:

$$\mathcal{L} = \sum_m w_m \mathcal{L}_m \; + \; \lambda \sum_m w_m \log w_m \tag{3}$$

Table 3 Representative multimodal and physiological emotion recognition studies in HRC

Study	Fusion method/Modalities	Robot platform	Real-time	Accuracy (if reported)	Highlights/Challenges
[23]	Hybrid fusion of speech + vision	Pepper	Yes	+12.5% vs unimodal	Real-time HRC; improved recognition in ambiguous dialogue
[22]	Late fusion of facial + context features	Elderly-care robot	Yes	N/A	Adaptive dialogue; emotion-aware task planning
[18]	EEG + behavioral cues	Custom platform	Partially	78.3% (lab)	Motion artifacts, calibration issues; promising internal affect mapping
[19]	EEG + GSR + facial expressions	N/A	No	81.2% (offline)	Early integration of wearables; proof-of-concept only
[28]	HRV + visual + speech	N/A	Partially	84.5% (simulated)	Emphasizes physiological-speech synergy; limited in-field tests
[42]	Multimodal memory network (text + vision + audio)	NA	No	82.1%	Pioneering architecture; not designed for robotics

Table 4 Comprehensive summary of emotion recognition in HRI/HRC (2015–2025)

Study	Modality	Model/Method	Features	Dataset	Robot	Performance
[59]	Speech	LadderNet + Beamforming approach for distant speech emotion recognition	CCC, far-field audio processing, robot ego-noise handling	Simulated + Room impulse responses	PR2	+180% CCC improvement over baseline methods
[13]	Facial	CNN-LSTM Hybrid architecture for real-time facial emotion detection	Facial landmarks detection and temporal modeling	CK+, FER2013	NAO	Accuracy: CK+ –96.3%, FER2013 –68.8%
[56]	Multimodal	ResNet-50 (video), Conv1D (speech) with late fusion strategy	Spectrograms + facial features extraction and combination	SAVEE, RAVDESS	NAO	Speech: 70%, Facial: 63.5%. Fusion improves accuracy
[48]	Speech	Vision Transformer/ BEiT transformers for personalized emotion recognition	Mel-spectrograms with attention mechanisms	User-specific speech data	NAO	SOTA performance per individual user
[60]	Facial	Convolutional Neural Network for facial expression analysis	Facial expressions analysis with geometric features	CK+, BU4DFE, JAFFE	NAO	Human-level performance with 29 users
[20]	Facial	DeepFace CNN for engagement-sensitive emotion recognition	Facial attributes extraction for engagement assessment	In-situ live HRI scenarios	NAO	Qualitatively good performance in quiz game
[61]	Facial	CNN Ensemble combining EfficientNet and VGG19 architectures	Multi-CNN pooled features with transfer learning	FER2013, CK+, HER	Custom humanoid	Accuracy: HER - 97%, FER2013 - 94.4%
[45]	Speech	LadderNet + wav2vec for enhanced far-field speech processing	Far-field enhanced speech with self-supervised learning	Room-acoustic simulated	NAO	+15% CCC vs MVDR baseline
[18]	EEG	SVM, Random Forest, MLP for brain signal classification	Frontal brain asymmetry analysis for valence detection	EEG signals (Emotiv)	Pepper	Valence/ arousal: up to 92% accuracy
[62]	Physiological	Deep neural networks for multi-signal physiological analysis	Bio-signal time-series (ECG, GSR, respiration)	AFFECT-HRI dataset	Anthropomorphic robot	Multi-user dataset validation
[63]	Body Pose	LSTM + Joint transformations for body motion analysis	Body motion analysis from skeletal joint data	Emotional Body Motion DB	Pepper	PCC = 0.81 for emotional intensity
[57]	Multimodal	Cross-channel CNN for face and body integration	Facial expressions and posture features fusion	In-house, live iCub	iCub	Accuracy: 74% in live deployment
[58]	Multimodal	GA-optimized Neural Network + HMM for adaptive behavior	Fusion-based speech + visual feature integration	Custom HRI data	Bioloid	Real-time adaptation: 97% accuracy
[64]	Multimodal	ResNet + DNN fusion for child-robot interaction	Facial and body posture features with hierarchical processing	Child-robot HRI (Zeno/ Furhat)	Furhat/ Zeno	Hierarchical fusion: 72% accuracy
[65]	Physiological	CNN processing EEG + GSR + Blood Volume Pulse signals	Brain signals + bio-signals multimodal integration	HRI with pepper	Pepper	Valence-arousal: > 80% accuracy

Table 4 (continued)

Study	Modality	Model/Method	Features	Dataset	Robot	Performance
[66]	Physiological	Survey of ECG, EEG, GSR methods for HRI applications	Trust and emotion bio-signals analysis techniques	Multiple reviewed datasets	Various	Survey reports: 85% average
[67]	Multimodal	Hybrid speech + vision fusion for dialogue enhancement	Speech prosody + visual features for dialogue context	Custom dialogue dataset	Pepper	+12.5% vs unimodal approaches
[22]	Multimodal	Late fusion facial + audio for elderly care applications	Facial + audio feature fusion for elderly-specific needs	Elderly-care dataset	Elderly-care robot	Adaptive dialogue system

where $\mathcal{L}_m$ is the per-modality task loss and $\lambda \geq 0$ governs regularisation strength [23]. For attention-based fusion, the loss takes the same form but weights are learned via cross-attention rather than quality signals, with L2 weight decay on the projection matrices as the regulariser to mitigate overfitting on small HRI and HRC corpora [42].

Table 2 summarized the challenges faced in different literature for multimodal ER. Ethical note: Physiological signals such as EEG and HRV are biometric by nature; their collection in HRC settings requires explicit informed consent and data minimisation protocols.

5.2 Emerging trends and opportunities

Multimodal emotion recognition is rapidly evolving within the context of HRC. Emerging research emphasizes embedding affective models into the robot's decision-making loop, allowing for emotion-aware task planning, adaptive feedback, and proactive social engagement [22]. As real-world HRI scenarios become more complex and user-centric, multimodal perception is expected to play a key role in enabling emotionally responsive interactions.

Future directions include:

- Cross-modal alignment via self-supervised learning: Leveraging contrastive objectives to link latent emotional representations across modalities.
- Wearable physiological sensors: Integration of mobile biosensing devices into robotic ecosystems for continuous affect monitoring.
- Standardized datasets and benchmarks: Developing unified corpora with synchronized emotion labels across valence-arousal and categorical spaces for HRC tasks.

By combining behavioral and physiological signals, multimodal systems can make HRC more intuitive, context-aware, and emotionally intelligent. A representative set of multimodal studies is summarized in Table 3.

A detailed summary table of emotion recognition studies in HRI and HRC from 2015 to 2025 is presented in Table 4. It covers a range of input modalities, including facial expressions, speech, body posture, and physiological signals, along with the models used, datasets, and robot platforms. The studies reflect diverse approaches, from classical classifiers to deep learning models, and highlight trends toward multimodal integration and real-world deployment. While the review classifies the studies according to modalities, real-world deployment, especially in HRC settings, involves additional

practical considerations such as cost, user acceptance, and integration complexity. Due to the scope of this article, a detailed analysis of these metrics was not conducted. Additionally, though we summarize multimodal fusion strategies across the reviewed studies, a deeper or more rigorous technical comparison such as performance trade-offs between early, late, and attention-based fusion was beyond the current scope. Hence, these aspects and discussions for multi modal methods are acknowledged as important for future work.

6 State-of-the-art emotion recognition in HRC

Building upon the modality-wise review presented in the previous sections, we now offer a chronological perspective to contextualize the field's trajectory. While earlier sections categorized approaches by input modality, this roadmap highlights how emotion recognition in HRC has evolved over time, marking technical milestones, shifting research priorities, and deployment advances, especially leading empirical contributions.

6.1 Early years (2015-2018): emergence of deep learning for affective HRI

The period between 2015 and 2018 marked a transition from handcrafted features to deep learning. Barros et al. [68] introduced early convolutional architectures (CCCNN) for spontaneous facial expression recognition in real-time robot settings. Around the same time, Lakomkin et al. [69] demonstrated robustness improvements in speech emotion recognition under realistic ego-noise by leveraging noise-aware data augmentation on the iCub platform. These early works emphasized the feasibility of deploying affective models in unconstrained environments, albeit still largely unimodal.

6.2 2019-2020: fusion architectures and physiological integration

By 2019, researchers began exploring cross-modal learning and user-centered features. Filntisis et al. [64] combined facial and posture cues, while Yu and Tapus [70] introduced thermal imaging and gait for adaptive interaction. In 2020, Val-Calvo et al. [71]

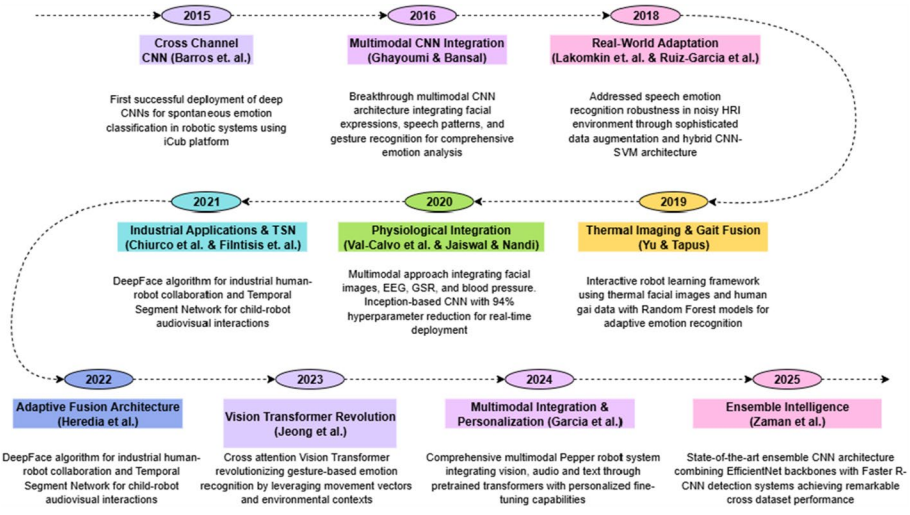


Fig. 6 Timeline of representative state-of-the-art contributions in emotion recognition for HRC (2015-2025), showing the evolution from early CNNs to personalized multimodal systems. Key external AI milestones contextualising HRC adoption are Transformer (2017), BERT (2018), wav2vec 2.0 (2020) and ViT (2021)

integrated facial features with biosignals (EEG, GSR) in real-time HRI scenarios, highlighting growing interest in dimensional emotion modeling and embodied physiological sensing.

6.3 2021-2023: real-time constraints and transformer-based models

As robot integration matured, real-time constraints, edge optimization, and transformer-based models became central. Zhang [72] optimized FER pipelines for industrial robots using DeepFace. Concurrently, Heredia et al. [73] and Filntisis et al. [74] employed fusion techniques that adaptively weighted modalities, addressing partial data or noisy environments. Grágeda et al. [45] tackled distant SER using beamforming and realistic acoustic simulations, showcasing deployment-ready architectures.

6.4 2024-2025: personalization and ensemble intelligence

Recent work reflects a shift toward personalized, multimodal, and ensemble systems tailored for diverse user behaviors. Garcia et al. [67] deployed a multimodal Pepper-based system combining vision, audio, and text, fine-tuned to individual users using pretrained transformer backbones. Zaman et al. [61] leveraged ensemble CNNs (EfficientNet, Faster R-CNN, VGG19) for FER in robot-perspective images, achieving SOTA accuracies on custom HRC datasets. These works represent a mature phase in the field, balancing high accuracy with practical HRI integration.

6.5 Synthesis and emerging trends

As depicted in Fig. 6, the research landscape has steadily evolved:

- From Unimodal to Multimodal: Vision and speech dominated early systems, but post-2020 work embraces multimodal fusion with physiological and contextual inputs.
- From Controlled Labs to Real-World Deployment: Advances in data augmentation, noise-robust architectures, and embedded optimization have improved real-world applicability.
- From Static Models to Personalized Systems: The field has moved toward user-specific adaptation, using fine-tuning, transfer learning, and transformer architectures to handle variability.

7 Future research directions

Despite notable advances in emotion-aware HRC, substantial gaps remain in ensuring system robustness, adaptability, and ethical deployment in unconstrained environments. Drawing from our synthesis of multimodal studies between 2015 and 2025, this section outlines key future research directions essential for bridging laboratory performance and real-world applicability.

7.1 Robust domain adaptation and generalization

A recurring limitation across surveyed literature was poor generalization to real-world environments. Emotion recognition models often suffer performance degradation when transitioning across deployment settings due to variations in acoustic conditions, sensor placement, or user demographics. Fewer than 20% of studies explicitly evaluated models across diverse environmental or cross-device settings.

Future efforts should prioritize:

- Domain-Invariant Learning: Techniques such as adversarial domain adaptation and contrastive learning can help extract features resilient to environment-specific shifts [75].
- Unsupervised and Test-Time Adaptation: Minimal supervision approaches (e.g., self-training, entropy minimization) offer promising results in adapting to unseen conditions [76].
- Cross-Robot Generalization: Ensuring that models trained on one platform (e.g., NAO) generalize to others like Pepper or PR2 remains a key challenge [16].

7.2 Multimodal integration and alignment

While over 70% of studies focused on unimodal pipelines, fusion strategies remain underutilized despite their theoretical robustness. Future work can explore:

- Temporal Alignment Strategies: Attention-based alignment and sequence modeling techniques (e.g., transformers, dynamic time warping) can mitigate synchronization issues across modalities.
- Uncertainty-Aware Fusion: Runtime modality weighting based on signal quality (e.g., SNR, face occlusion) improves resilience under partial input degradation [23].
- Hierarchical and Modular Fusion: Architectures that combine low-level signal fusion with high-level decision aggregation offer interpretability and scalability [22, 77].
- Diffusion-based synthesis, as demonstrated by CP-Diffusion [78], offers a promising route to augmenting scarce HRC training data with realistic, prompt-conditioned emotional video sequences.

7.3 Personalization and culturally adaptive modeling

Only a small fraction (10-15%) of existing systems incorporate user-specific or culturally aware modeling, despite the subjectivity of emotion expression. Moving forward, future systems must embed adaptive frameworks that reflect user diversity and dynamic emotional baselines. Key areas include:

- Few-Shot and Meta-Learning: Learning from a few labeled samples per user can enable rapid personalization with minimal burden [79].
- Continual Learning: Emotion recognition systems should incrementally update models over time while avoiding catastrophic forgetting.
- Cross-Cultural Datasets: Broader inclusion of emotion corpora across cultural contexts is necessary to reduce model bias and ensure inclusivity.

7.4 Real-time and edge-aware system design

Robots deployed in homes, schools, or healthcare settings must operate with tight compute and latency constraints. Future research should emphasize:

- Model Compression: Techniques such as pruning, quantization, and knowledge distillation can help reduce inference time and memory footprint [49].
- On-Device Learning: Edge-focused adaptation (e.g., federated learning or local fine-tuning) can enable continual updates without central data sharing.

These strategies are essential to move beyond cloud-reliant pipelines and bring scalable emotion understanding to low-power, embedded robotic platforms.

7.5 Ethical, privacy, and societal considerations

As emotion-aware systems become more pervasive, they raise critical concerns around surveillance, emotional manipulation, and algorithmic fairness. Among the studies reviewed, fewer than 10% explicitly addressed privacy or fairness. Future frameworks must prioritize:

- Privacy-Preserving Learning: Federated or differentially private learning approaches can mitigate data leakage and ensure user trust [55].
- Fairness Auditing: Systems must be evaluated across gender, age, and cultural subgroups to identify and mitigate latent biases.
- Transparency and Explainability: Emotion-based decisions in robots should be interpretable, particularly in sensitive domains like elder care or education.

In sum, future research in HRC emotion recognition must adopt a multi-pronged approach, technically robust, user-adaptive, resource-aware, and ethically grounded, to build systems that are not only intelligent but also socially responsible.

8 Conclusion

This literature review synthesizes recent developments in emotion recognition for HRI and HRC, spanning studies published between 2015 and 2025. Our analysis reveals a clear evolution toward deep learning architectures, hybrid modeling strategies and signal processing-aware designs across speech, facial, and multimodal modalities. SER has advanced through the integration of attention mechanisms, spatial filtering, and self-supervised learning. These approaches have improved robustness under real-world far-field and noisy conditions, aligning well with the acoustic challenges of voice-enabled robotic platforms. In contrast, FER systems are highly effective in controlled settings, it remains limited by occlusion, lighting variation, and camera placement, factors frequently encountered in real-world HRC. Multimodal and physiological approaches, such as the fusion of speech, facial, and EEG signals, show promise in enhancing emotion recognition through complementary cues. However, their deployment is hindered by synchronization issues, hardware constraints, and the scarcity of synchronized multimodal datasets tailored for HRC.

A consistent theme across modalities is the 4-27.5% drop in performance when transitioning from laboratory to operational environments, with speech systems averaging 8.6% and facial systems 14.4% degradation across paired studies. This gap underscores the need for more robust domain adaptation techniques, personalization strategies, and lightweight, real-time deployment frameworks. Looking forward, research must increasingly focus on developing emotion recognition systems that are generalizable, computationally efficient, and ethically sound. Priorities include cross-platform transfer learning, privacy-preserving computation, culturally adaptive modeling, and explainable, on-device inference. Emotion-aware robots have the potential to transform domains such as healthcare, education, and assistive living, but their societal acceptance will depend on building systems that are not only intelligent, but also inclusive, transparent, and respectful of human values.

In conclusion, this work highlights not only the technical progress made in emotion-aware HRC but also the pressing need for adaptive, multimodal, and ethically aligned research moving forward.

Acknowledgements
This work was partially supported by PNRR projects FAIR-Future AI Research (Grant No. PE00000013) and INTAFAM00039 MUSIC4D.

Author contributions
Nikhilsingh Parihar—Led the core technical work for the study, including analyzing the research domain, reviewing and interpreting the selected literature, synthesizing key findings across modalities, and contributing to structuring and refining the manuscript. Also supported data extraction and result interpretation throughout the review process. Rashmi Chawla—Designed and executed the strategic literature review methodology. Responsible for defining the search strategy, developing the PRISMA-inspired screening process, organizing modality-wise classification (speech, facial, multimodal, physiological), and drafting major sections of the manuscript. Ensured methodological rigor and coherence across the review. Giancarlo Fortino-Provided overall supervision and expert guidance throughout the project. Contributed to validating the review framework, refining the thematic structure, and ensuring scientific accuracy. Reviewed the full manuscript, offered critical revisions, and approved the final version.

Funding
This research received no external funding.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This article does not contain any studies with human participants or animals performed by any of the authors. Ethics declaration: not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 24 November 2025 / Accepted: 15 April 2026
Published online: 28 April 2026

References

1. Cowie R, Douglas-Cowie E, Savvidou S, McMahon E, Sawey M, Schröder M. Emotion: the missing link? IEEE Trans Syst Man Cybern A. 2001;10(1):38–48.
2. Ekman P. An argument for basic emotions. Cogn Emot. 1992;6(3–4):169–200.
3. Scherer KR. What are emotions? And how can they be measured? Soc Sci Inf. 2005;44(4):695–729.
4. Picard RW. Affective computing: challenges. Int J Hum-Comput Stud. 2003;59(1–2):55–64.
5. Breazeal C. Emotion and sociable humanoid robots. Int J Hum-Comput Stud. 2003;59(1–2):119–55.
6. Baraka K, Rosenthal S, Veloso M. Enhancing human understanding of a mobile robot's state and actions using expressive lights. In Proc. 25th IEEE Int. Symp. Robot Hum. Interact. Commun. (RO-MAN), 2016;652–657.
7. Paiva A, Leite I, Boukricha H, Wachsmuth I. Empathy in virtual agents and robots: a survey. ACM Trans Interact Intell Syst. 2017;7(3):11:1-11:40.
8. Yu C, Mathur A, Coaley S, Scuteri P, Rivas M, et al. Socially assistive robots for people with dementia: systematic review and meta-analysis of feasibility, acceptability and the effect on cognition, neuropsychiatric symptoms and quality of life. Ageing Res Rev. 2022;78:101633.
9. Devillers L, Vidrascu L, Lamel L. Real-life emotions detection with lexical and paralinguistic cues on human-human call center dialogs. In Proc. Interspeech, 2005;549–552.
10. Zhang J, Yin Z, Chen P, Nichele S. Emotion recognition using multi-modal data and machine learning techniques: a tutorial and review. Inf Fusion. 2020;59:103–26.
11. Scassellati B, Admoni H, Mataric M. Robots for use in autism research. Annu Rev Biomed Eng. 2012;14:275–94.
12. Kennedy J, Baxter P, Senft E, Belpaeme T. Higher nonverbal immediacy leads to greater learning gains in child-robot tutoring interactions. In Proc. 7th Int. Conf. Social Robotics, 9388 of Lecture Notes in Computer Science, 2015;327–336.
13. Li TH, Kuo PH, Tsai TN, Luan PC. Cnn and lstm based facial expression analysis model for a humanoid robot. IEEE Access. 2019;7:93998–4011.
14. Ruiz-Garcia A, Elshaw M, Altahhan A, Palade V. A hybrid deep learning neural approach for emotion recognition from facial expressions for socially assistive robots. Neural Comput Appl. 2018;29(7):359–73. https://doi.org/10.1007/S00521-018-3358-8
15. R. Chawla, R. Longo, F. Pupo and G. Fortino, "EAC-HRC: Emotion-Aware Communication for Human-Robot Collaboration in a Music Orchestra System," 2025 IEEE International Conference on Cyber Humanities (IEEE-CH), Florence, Italy, 2025, pp. 1-6. https://doi.org/10.1109/IEEE-CH65308.2025.11279369

16. Grágeda N, Busso C, Alvarado E, García R, Mahu R, Huenupán F, et al. Speech emotion recognition in real static and dynamic human-robot interaction scenarios. Comput Speech Lang. 2024;84:101567.
17. Noroozi F, Corneanu C, Kaminska D, Sapinski T, Escalera S, Anbarjafari G. Survey on emotional body gesture recognition. IEEE Trans Affect Comput. 2021;12(2):505–23.
18. Staffa M, D'Errico L. Eeg-based machine learning models for emotion recognition in hri. Interacción. 2023;14(1):1–8.
19. Tripathi S, Acharya S, Sharma RD, Mittal S, Bhattacharya S. Using deep and convolutional neural networks for accurate emotion classification on deap dataset. In Proc. 31st AAAI Conf. Artif. Intell., 2017;4746–4752. https://doi.org/10.1609/aaai.v31i2.19105
20. Chiurco A, Frangella J, Longo F, Nicoletti L, Padovano A, Solina V, Mirabelli G, Citraro C. Real-time detection of worker's emotions for advanced human-robot interaction during collaborative tasks in smart factories. In Proc. IEEE Int. Symp. Multimedia (ISM), 2021;1–6. https://doi.org/10.1016/J.PROCS.2022.01.388
21. Poria S, Cambria E, Bajpai R, Hussain A. A review of affective computing: from unimodal analysis to multimodal fusion. Inf Fusion. 2017;37:98–125.
22. Fiorini L, Loizzo FGC, D'Onofrio G, Sorrentino A, Ciccone F, Russo S, Giuliani F, Sancarlo D, Cavallo F. Can i feel you? Recognizing human's emotions during human–robot interaction. In Proc. 14th Int. Conf. Social Robotics (ICSR 2022), 13817 of Lecture Notes in Computer Science, 2023;507–517. https://doi.org/10.1007/978-3-031-24667-8_45
23. Garcia S, Gomez-Donoso F, Cazorla M. Enhancing human–robot interaction: development of multimodal robotic assistant for user emotion recognition. Appl Sci. 2022;12(24):12914.
24. Pant B, Sharma H, Chawla R, Kumar C. An iot-based intelligent traffic engagement system with emergency vehicles preemption. Int J Sens Netw. 2022;40(1):10–9.
25. Chawla R, Singhal P, Garg AK. Internet of things driven framework for smart solar energy system. J Energy Res Technol. 2020;142(1):011201. https://doi.org/10.1115/1.4044124
26. Russell JA. A circumplex model of affect. J Pers Soc Psychol. 1980;39(6):1161–78.
27. Kollias D, Tzirakis P, Nicolaou MA, Papaioannou A, Zhao G, Schuller B, et al. Deep affect prediction in-the-wild: Aff-wild database and challenge, deep architectures, and beyond. Int J Comput Vis. 2019;127(6–7):907–29.
28. Alskafi F,Khandoker A, Lee U, Park CY, Jelinek H. Multimodal analysis of physiological signals for wearable-based emotion recognition using machine learning. In Proc. Computing in Cardiology (CinC), 2022;1–4.
29. Zhao Z, Bao Z, Zhao Y, Zhang Z, Cummins N, Ren Z, et al. Exploring deep spectrum representations via attention-based recurrent and convolutional neural networks for speech emotion recognition. IEEE Access. 2019;7:97515–25.
30. Handrich S, Dinges L, Al-Hamadi A, Werner P, Saxen F, Aghbari Z. Simultaneous prediction of valence/arousal and emotion categories and its application in an hrc scenario. J Ambient Intell Humaniz Comput. 2021;12(1):2313–31.
31. Zhu X, Wang Y, Cambria E, Rida I, López JS, Cui L, et al. RMER-DT: Robust multimodal emotion recognition in conversational contexts based on diffusion and transformers. Inf Fusion. 2025;123:103268.
32. Wang, Rui & Xu, Duyun & Cascone, Lucia & Wang, Yaoyang & Chen, Hui & Zheng, Jianbo & Zhu, Xianxun. (2025). RAFT: Robust Adversarial Fusion Transformer for multimodal sentiment analysis. Array. 27. 100445. https://doi.org/10.1016/j.array.2025.100445
33. Zhu X, Liu Z, Cambria E, Yu X, Fan X, Chen H, et al. A client-server based recognition system: non-contact single/multiple emotional and behavioral state assessment methods. Comput Methods Programs Biomed. 2025;260:108564.
34. Wang R, Zhu J, Han S, Cambria E, Li Y, Kwok K. Contrastive-based removal of negative information in multimodal emotion analysis. Cogn Comput. 2025. https://doi.org/10.1007/s12559-025-10463-9.
35. Zhang Y, Chen H, Rida I, Zhu X, Cambria E. A generative random modality dropout framework for robust multimodal emotion recognition. IEEE Intell Syst. 2025;40:62–9.
36. Sun H, Zhang X, Liu W, Cambria E. Integrating audio-visual text generation with contrastive learning for enhanced multimodal emotion analysis. Inf Fusion. 2025. https://doi.org/10.1016/j.inffus.2025.103809.
37. Zhu X, Guo C, Feng H, Wang R. A review of key technologies for emotion analysis using multimodal information. Cogn Comput. 2024;16:1504–30.
38. Zhu X, Feng H, Guo C, Fan X, Huang J, Wang R. EMVAS: end-to-end multimodal emotion visualization analysis system. Complex Intell Syst. 2025;11:374.
39. Wang R, Zhu J, Wang S, Wang T, Huang J, Zhu X. Multi-modal emotion recognition using tensor decomposition fusion and self-supervised multi-tasking. Int J Multimed Inf Retr. 2024;13:39.
40. Zhu X, Huang Y, Wang X, Wang R. Emotion recognition based on brain-like multimodal hierarchical perception. Multimed Tools Appl. 2024;83:56039–57.
41. LiuR, Yang K, Roitberg A, Stiefelhagen R. Knowledge generation and distillation for road segmentation in intelligent transportation systems. IEEE Trans Intell Transp Syst. 2025. PP. 1-13. https://doi.org/10.1109/TITS.2025.3577794
42. Zadeh A, Liang PP, Poria S, Cambria E, Morency LP. Memory fusion network for multi-view sequential learning. Proc AAAI Conf Artif Intell. 2018;32:5634–41.
43. Zhang D, Wu L, Li S, Zhu Q, Zhou G. Multi-modal language analysis with hierarchical interaction-level and selection-level attentions. In Proc. IEEE Int. Conf. Multimedia Expo (ICME), 2019;724–729.
44. Mustaqeem MS, Kwon S. Clustering based speech emotion recognition by incorporating learned features and deep bilstm. IEEE Access. 2020;8:79861–75.
45. Grágeda N, Alvarado E, Mahu R, Busso C, Yoma N. Distant speech emotion recognition in an indoor human-robot interaction scenario. In Proc. Interspeech, 2023;3657–3661.
46. García R, Mahu R, Grágeda N, Luzanto A, Bohmer N, Busso C, Yoma N. Speech emotion recognition with deep learning beamforming on a distant human-robot interaction scenario. In Proc. Interspeech, 2024;3215–3219.
47. Lakomkin E, Zamani MA, Weber C, Magg S, Wermter, S. On the robustness of speech emotion recognition for human-robot interaction with deep neural networks. In Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS), 2018;854–860.
48. R.Mishra, A. Frye, M. M. Rayguru and D. O. Popa, "Personalized Speech Emotion Recognition in Human-Robot Interaction Using Vision Transformers," in IEEE Robotics and Automation Letters, 2025;12(5):4890-4897. https://doi.org/10.1109/LRA.2025.3554949
49. Chen L, Su W, Feng Y, Wu M, She J, Hirota K. Two-layer fuzzy multiple random forest for speech emotion recognition in human-robot interaction. Inf Sci. 2020;509:150–63.

50. Melinte O, Vladareanu L. Facial expressions recognition for human–robot interaction using deep convolutional neural networks with rectified adam optimizer. Sensors. 2020;20(8):2393.
51. Jaiswal S, Nandi GC. Hyperparameters optimization for deep learning based emotion prediction for human robot interaction. 2020. https://doi.org/10.48550/arXiv.2001.03855
52. Minaee S, Abdolrashidi A. Deep-emotion: Facial expression recognition using attentional convolutional network. Sensors. 2019. https://doi.org/10.3390/s21093046.
53. Kollias D. Abaw: valence-arousal estimation, expression recognition, action unit detection & multi-task learning challenges. 2022.
54. Othmani A, Khan M, El Saddik A, Cogo Moreira H. ARNet: enhanced tracking and optimization for video-based affect recognition. In Proceedings of the 2025 7th International Conference on Image, Video and Signal Processing (IVSP), ACM, 2025.
55. Anwar MA, Agrawal M, Gahlan N, Sethia D, Singh G, Chaurasia R, et al. Fedemo: A privacy-preserving framework for emotion recognition using eeg physiological data. In Proc. 2023 14th Int. Conf. Commun. Syst. NETworkS (COMSNETS), 2023; 119–124. https://doi.org/10.1109/COMSNETS56262.2023.10041308
56. Jain S, Kumar N. Deep learning based emotion recognition in human–robot interaction using audio–visual fusion. AIP Conf Proc. 2024;3122:030007.
57. Hamester D, Barros P, Wermter S. Face expression recognition with a 2-channel convolutional neural network. In Proc. 2015 Int. Joint Conf. Neural Networks (IJCNN), 2015;1–8.
58. Perez-Gaspar LA, Caballero O, Trujillo-Romero F. Multimodal emotion recognition with evolutionary computation for human-robot interaction. Expert Syst Appl. 2016;66:53–61.
59. Grágeda M, Mahu R, Alvarado E, Yoma NB. Laddernet for far-field speech emotion recognition in human–robot interaction. IEEE Trans Affect Comput. 2023;14(4):2901–13.
60. Palmero-Pérez F, García-Aracil N, Azorín JM, Fernández E. A facial emotion recognition system for the nao robot using convolutional neural networks. Robot Auton Syst. 2020;128:103516.
61. Zaman MR, Samanta D, Das S. A novel emotion recognition system for human-robot interaction (hri) using deep ensemble classification. Int J Intell Syst. 2025. https://doi.org/10.1155/int/6611276.
62. Heinisch JS, Kirchhoff J, Busch P, Wendt J, von Stryk O, David K. The affect-hri data set: physiological data for affective computing in human-robot interaction with anthropomorphic service robots. Zenodo. 2024. https://doi.org/10.5281/zenodo.10422259.
63. Sun M, Mou Y, Xie H, Xia M, Wong M, Ma X. Estimating emotional intensity from body poses for human-robot interaction. 2019.
64. Filntisis P, Efthymiou N, Koutras P, Potamianos G, Maragos P. Fusing body posture with facial expressions for joint recognition of affect in child-robot interaction. EEE Robotics Autom Lett. 2019. https://doi.org/10.1109/LRA.2019.2930434.
65. Val-Calvo M, Álvarez Sánchez JR, Ferrández-Vicente JM, Díaz-Morcillo A, Fernández-Jover E. Real-time multi-modal estimation of dynamically evoked emotions using eeg, heart rate and galvanic skin response. Int J Neural Syst. 2020;30(4):2050013.
66. Savur C, Sahin F. Survey on physiological computing in human–robot collaboration. Machines. 2023;11(5):536.
67. Garcia S, Gomez-Donoso F, Cazorla M. Enhancing human–robot interaction: development of multimodal robotic assistant for user emotion recognition. Appl Sci. 2024;14(24):11914. https://doi.org/10.3390/app142411914
68. Barros P, Weber C, Wermter S. Emotional expression recognition with a cross-channel convolutional neural network for human-robot interaction, In Proc. 2015 IEEE-RAS 15th Int. Conf Humanoid Robots (Humanoids), 2015;582–587.
69. Lakomkin E, Weber C, Wermter S. On the robustness of speech emotion recognition for human-robot interaction with deep neural networks. In Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS), 2018;1228–1234.
70. Yu, Chuang, and Adriana Tapus. "Interactive robot learning for multimodal emotion recognition." In International conference on social robotics, pp. 633-642. Cham: Springer International Publishing, 2019. https://doi.org/10.1007/978-3-030-35888-4_59
71. Spezialetti M, Placidi G, Rossi S. Emotion recognition for human-robot interaction: recent advances and future perspectives. Front Robot AI. 2020;7:532279.
72. Zhang Z. Deep face emotion recognition. J Phys: Conf Ser. 2018;1087(6):062036.
73. Heredia J, Lopes-Silva E, Cardinale Y, Diaz-Amado J, Dongo I, Graterol W, et al. Adaptive multimodal emotion detection architecture for social robots. IEEE Access. 2022;10:20727–44.
74. Filntisis PP, Efthymiou N, Potamianos G, Maragos P. An audiovisual child emotion recognition system for child-robot interaction applications. In Proc. 29th Eur. Signal Process. Conf. (EUSIPCO), 2021;791–795.
75. Qi R, Zhao C, Guo X, Li Z, Li S, Chen H, et al. A cross-modal attention model with contextual enhancements for speech emotion recognition. Speech Commun. 2025;145:103285.
76. Zhao Y, Wang J, Lu C, Li S, Schuller B, Zong Y, Zheng W. Emotion-aware contrastive adaptation network for source-free cross-corpus speech emotion recognition. 2024.
77. Zhang Y, Cheng C, Zhang Y. Multimodal emotion recognition using a hierarchical fusion convolutional neural network. IEEE Access. 2021;9:7943–51.
78. Saeed M, Khan M, Saad M, Rahim N, Gueaieb W, El Saddik A. CP-diffusion: conditional prompt-based diffusion models for video generation. ACM Trans Multimed Comput Commun Appl. 2026. https://doi.org/10.1145/3793552.
79. R. Chawla, F. Pupo and G. Fortino, "Exploring Multi-Agent Human-Robot Teaming Dynamics in a Musical Orchestral Context," 2025 IEEE 5th International Conference on Human-Machine Systems (ICHMS), Abu Dhabi, United Arab Emirates, 2025, pp. 356-361 https://doi.org/10.1109/ICHMS65439.2025.11154360

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.